

Bluestock Mutual Fund Analytics

Final report

A reproducible analytics workflow covering ETL, SQLite, performance measurement, investor behavior, and a four-page interactive dashboard.

Executive Summary

The model contains 40 schemes, 32,778 investor transactions, and daily NAV observations for performance analysis.

The strongest one-year return in the supplied scheme table is 24.93% for ABSL Small Cap Fund - Regular - Growth.

The largest observed fund-house AUM is associated with SBI Mutual Fund. The dashboard turns these results into four decision views.

Data Sources

Source files cover fund master data, NAV history, fund-house AUM, monthly SIP inflows, category inflows, folios, scheme performance, investor transactions, portfolio holdings, and benchmark indices.

All source paths are relative to the repository and are transformed into data/processed by the ETL pipeline.

ETL Design

`scripts/etl_pipeline.py` validates required inputs, normalizes column names, parses date fields, removes duplicates, handles infinite numeric values, and writes deterministic cleaned CSVs.

`src/load_db.py` loads every cleaned table into SQLite and creates indexes for NAV and transaction lookups. The master runner executes the workflow in order.

EDA Findings: Industry

AUM trends are aggregated by date and fund house. The dashboard highlights industry scale, AMC concentration, SIP inflows, and folio growth.

Units are kept explicit: lakh crore for industry AUM, crore for scheme AUM and flows, and crore folios for investor reach.

EDA Findings: Investors

Investor transactions are segmented by state, transaction type, age group, city tier, and month.

The most useful operational comparisons are transaction value mix, SIP amount by age group, geographic concentration, and monthly transaction volume.

Performance Analysis

Daily NAV returns are annualized with 252 trading days. Sharpe uses a 6% annual risk-free rate. Beta uses covariance with the NIFTY50 series divided by benchmark variance.

Maximum drawdown is calculated from the cumulative return curve. Historical 95% daily VaR is the 5th percentile of daily returns and is reported as a negative threshold.

Performance Results

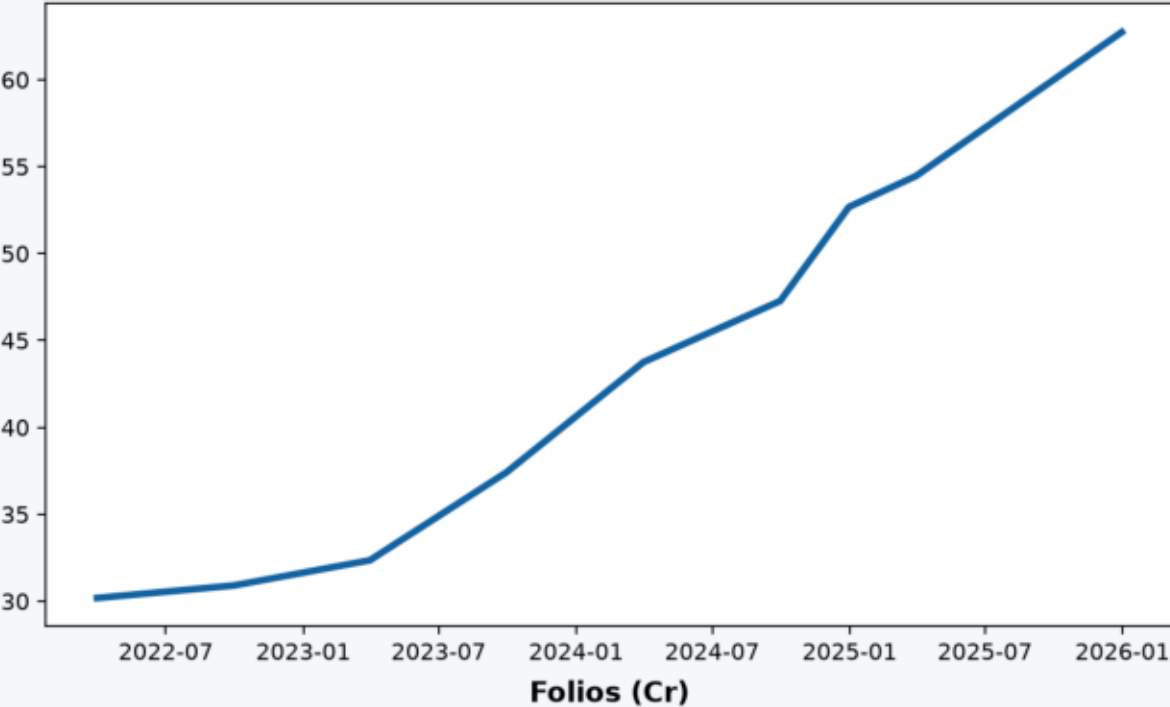
The metrics output contains 40 funds and 1149 to 1149 daily return observations per fund.

The scorecard combines return, Sharpe ratio, volatility, rating, and risk grade. The recommendation logic is intentionally explainable and should not be treated as investment advice.

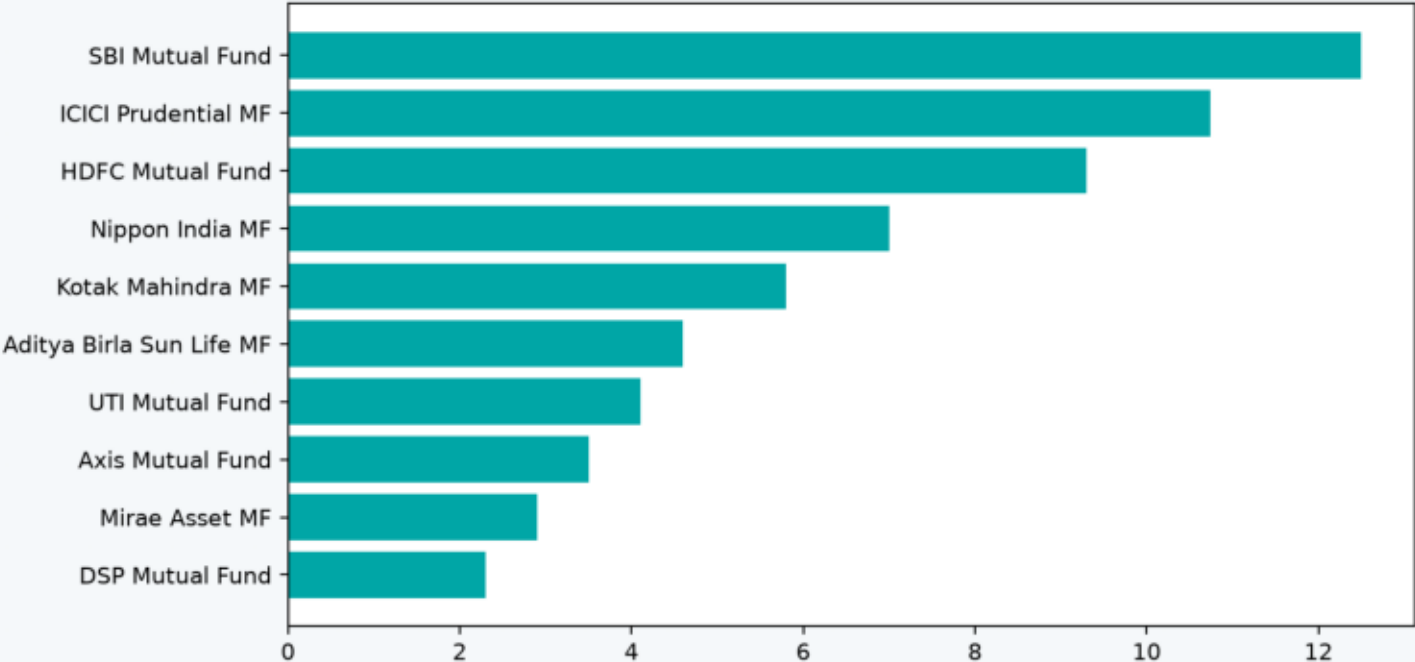
Bluestock Mutual Fund Analytics | Industry Overview

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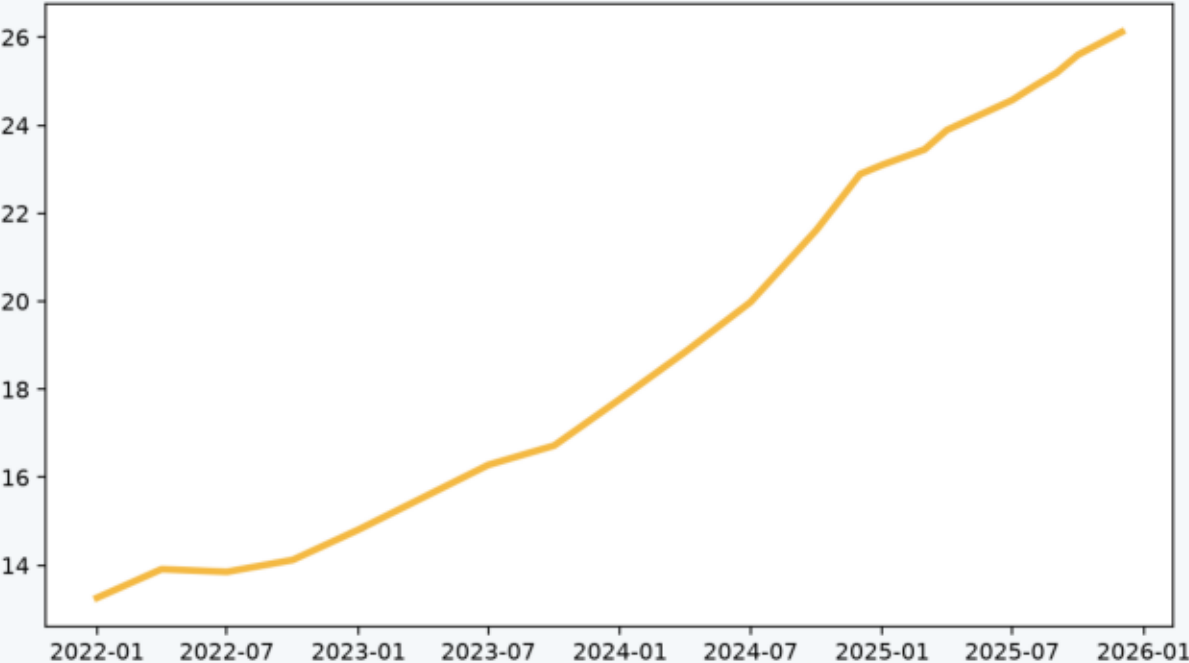
Industry AUM trend



AUM by AMC



Folios (Cr)



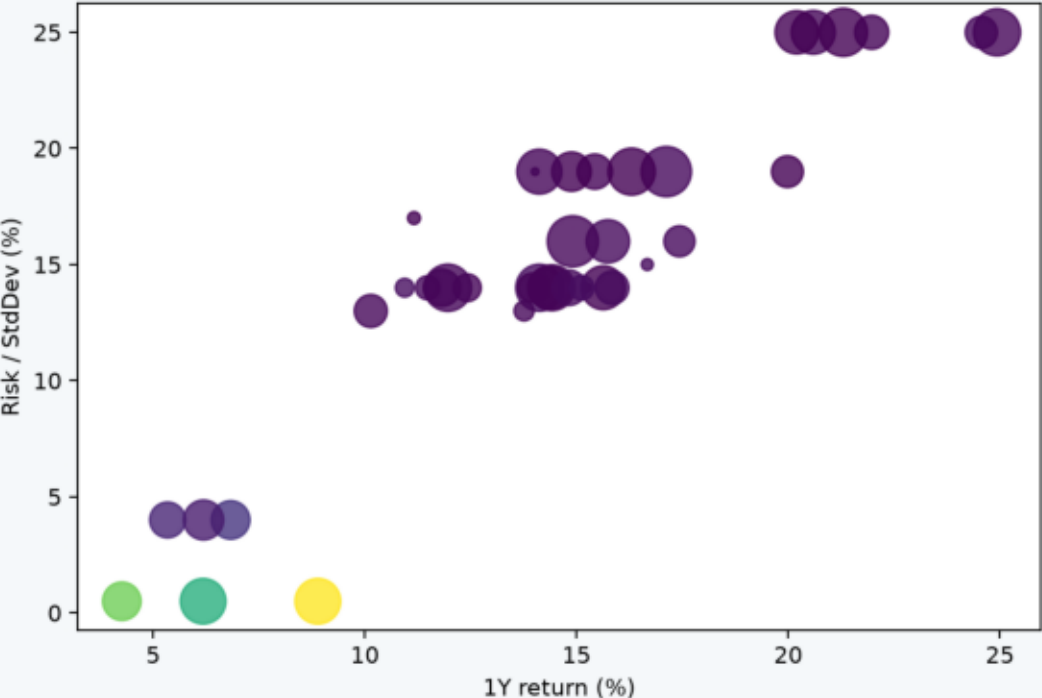
₹81L Cr
Total AUM

₹31K Cr SIP | 26.12 Cr Folios | 1,908 Schemes

Bluestock Mutual Fund Analytics | Fund Performance

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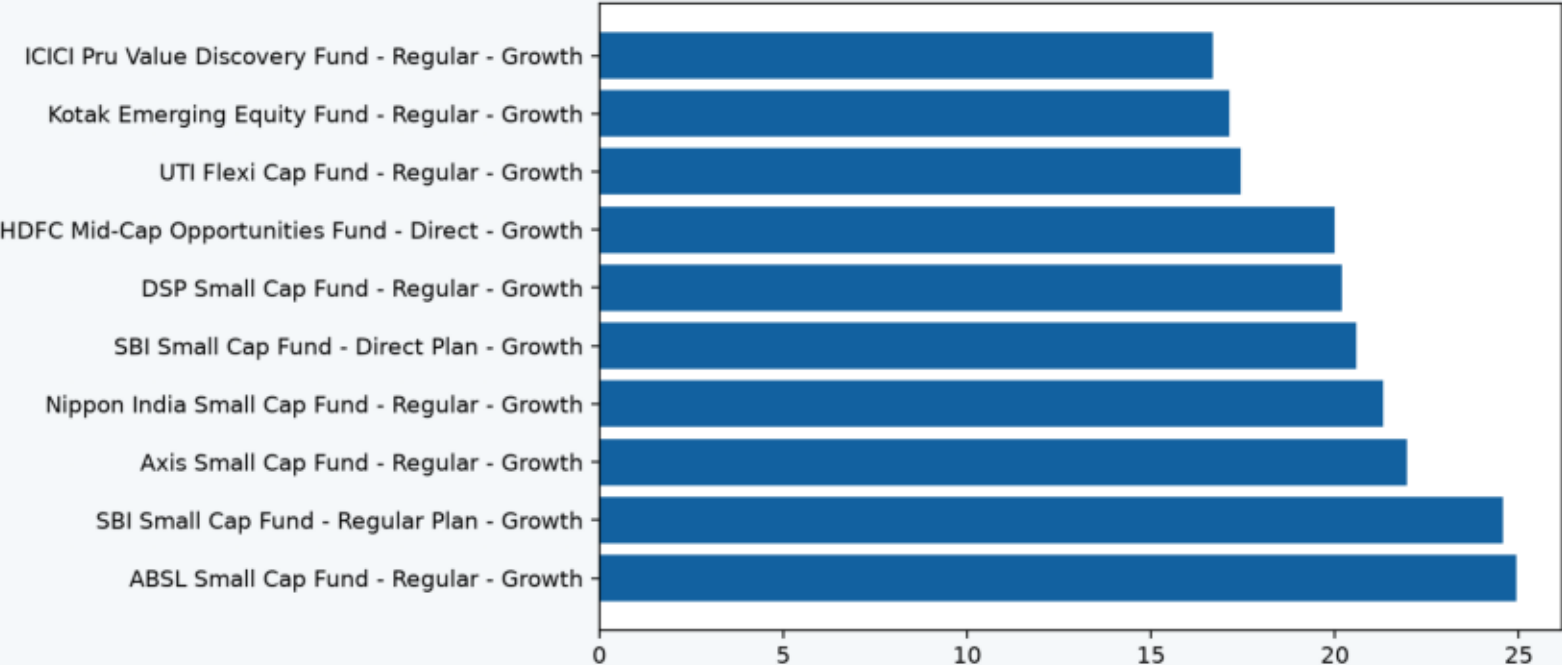
Return vs risk



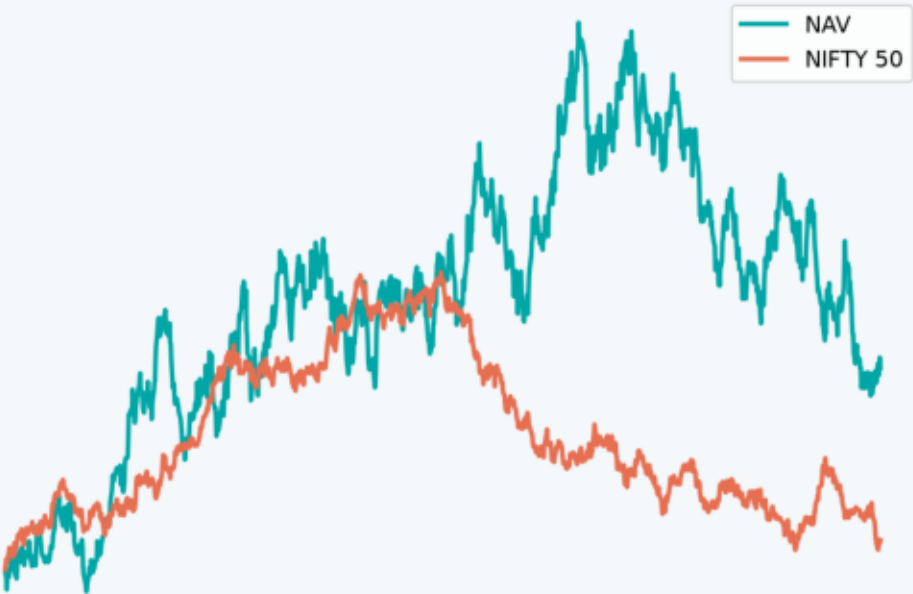
Fund scorecard (top 8)

scheme_name	return_1yr_pct	sharpe_ratio
ABSL Small Cap Fund - Regular - Growth	24.9%	0.90
SBI Small Cap Fund - Regular Plan - Growth	24.6%	0.94
Axis Small Cap Fund - Regular - Growth	22.0%	0.84
Nippon India Small Cap Fund - Regular - Growth	21.3%	0.81
SBI Small Cap Fund - Direct Plan - Growth	20.6%	0.93
DSP Small Cap Fund - Regular - Growth	20.2%	0.80
HDFC Mid-Cap Opportunities Fund - Direct - Growth	20.0%	0.80
UTI Flexi Cap Fund - Regular - Growth	17.4%	0.96

Top funds by 1Y return

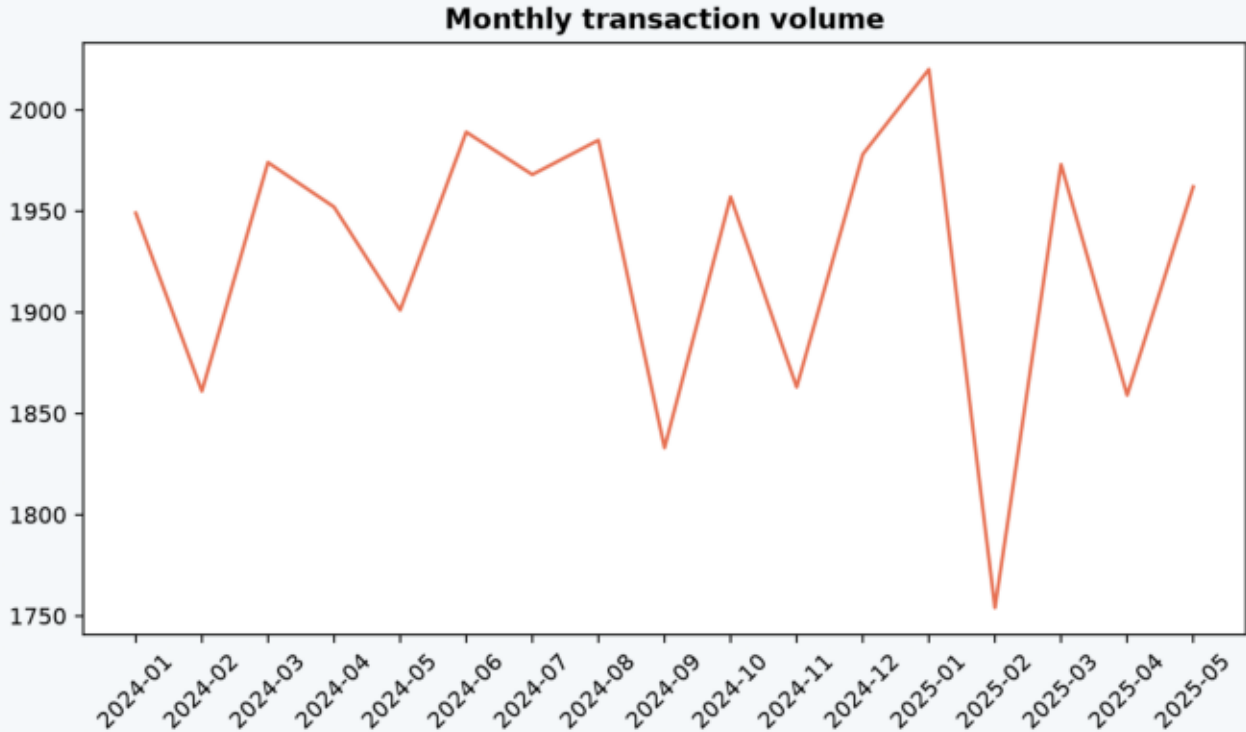
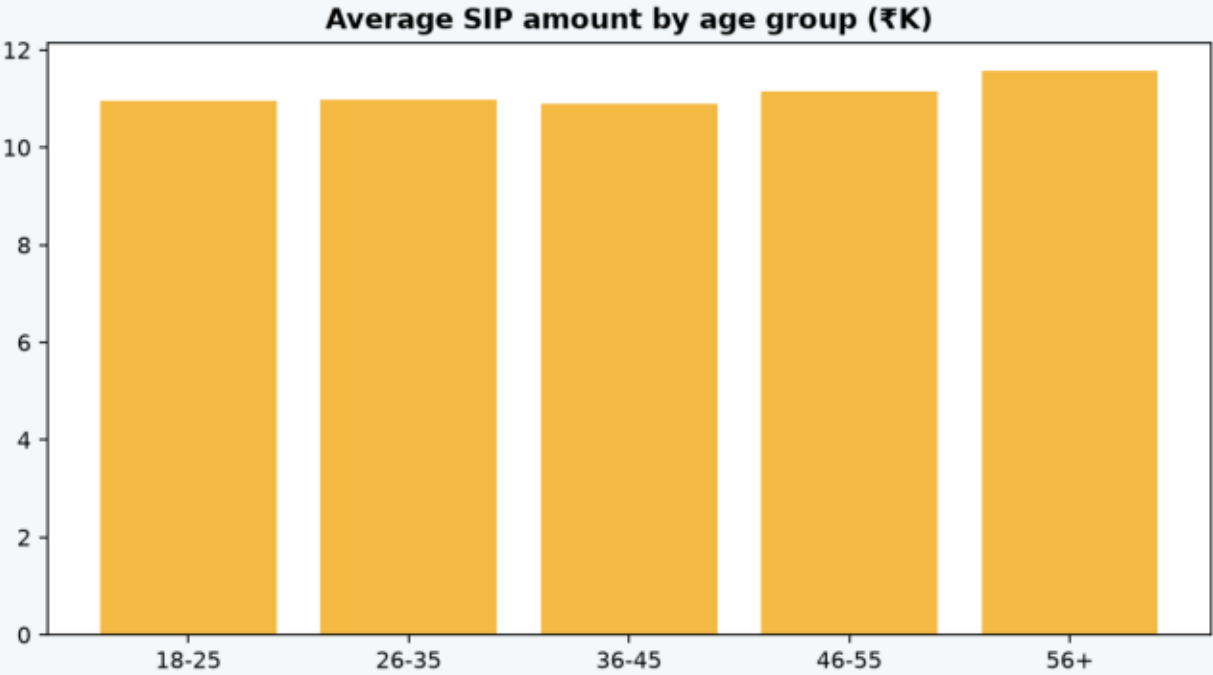
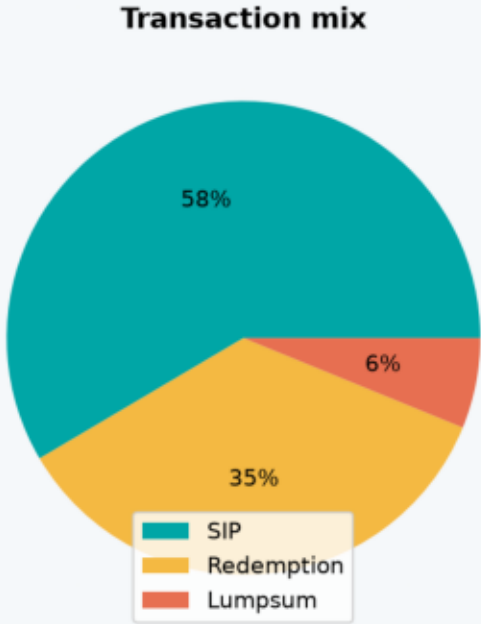
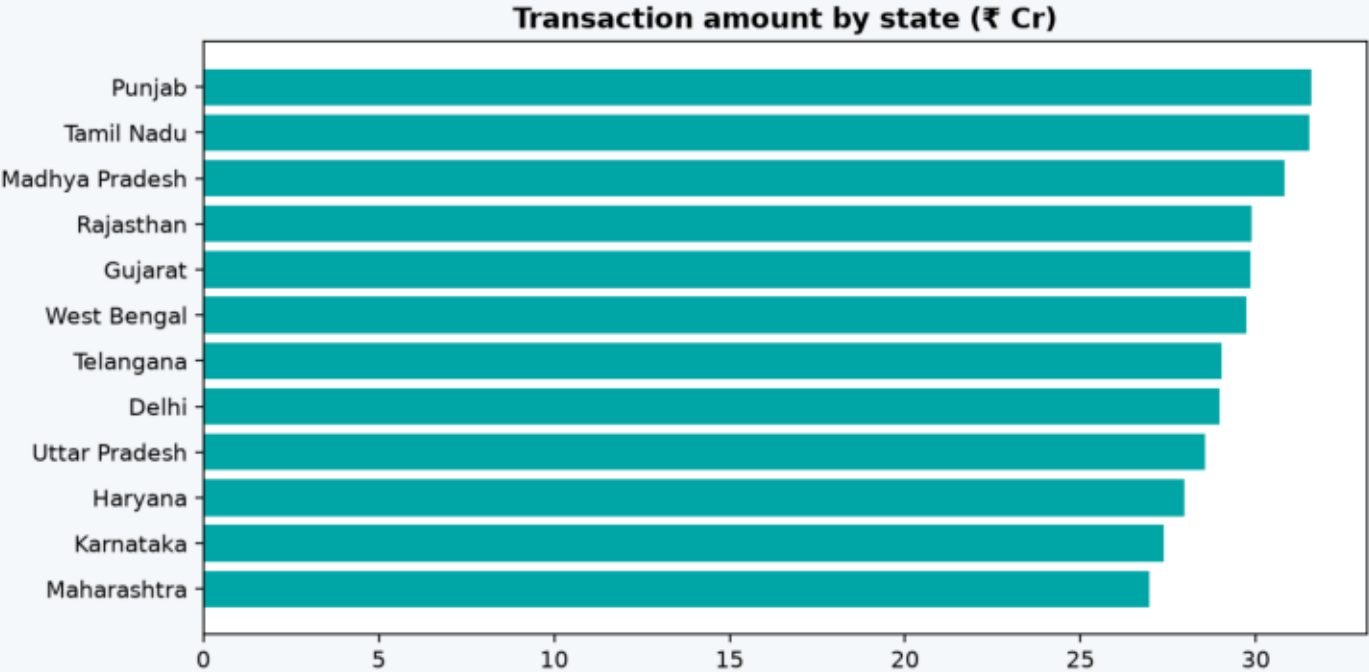


NAV vs benchmark (rebased to 100)



Bluestock Mutual Fund Analytics | Investor Analytics

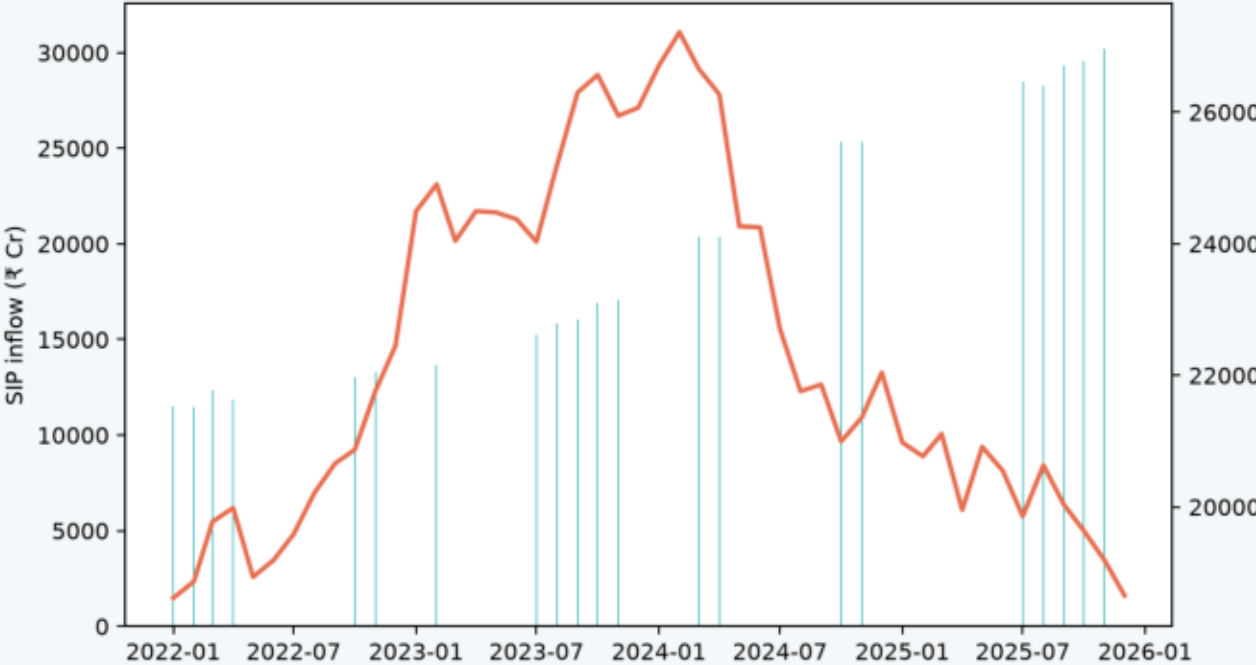
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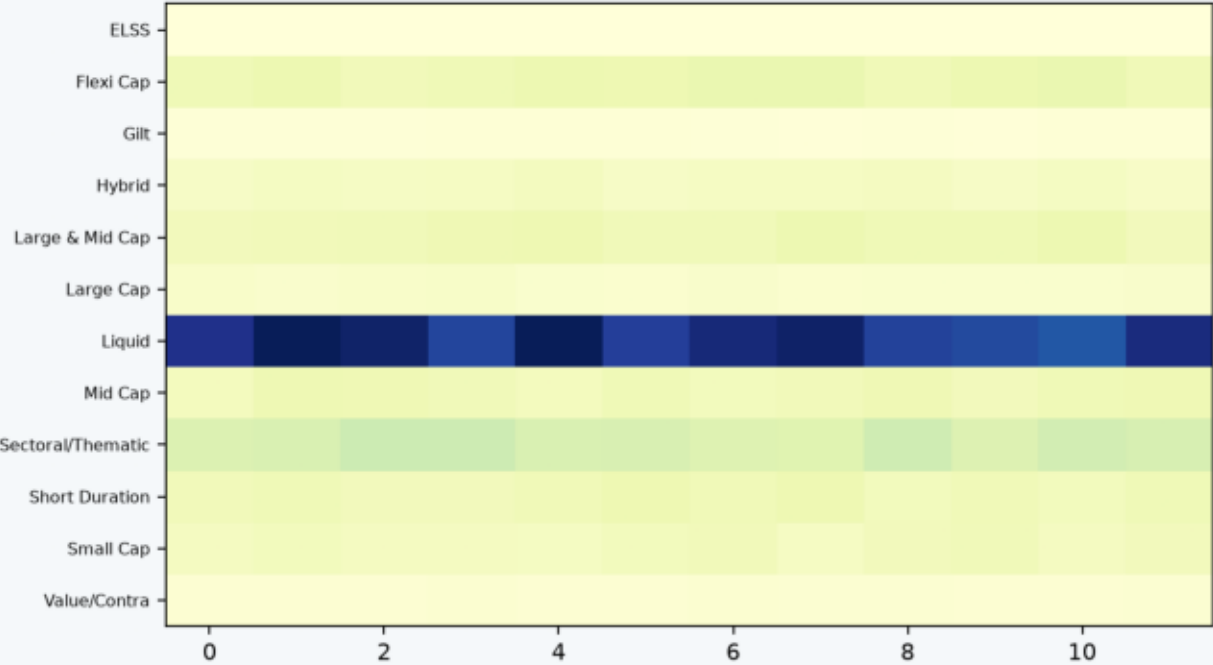
Bluestock Mutual Fund Analytics | SIP and Market Trends

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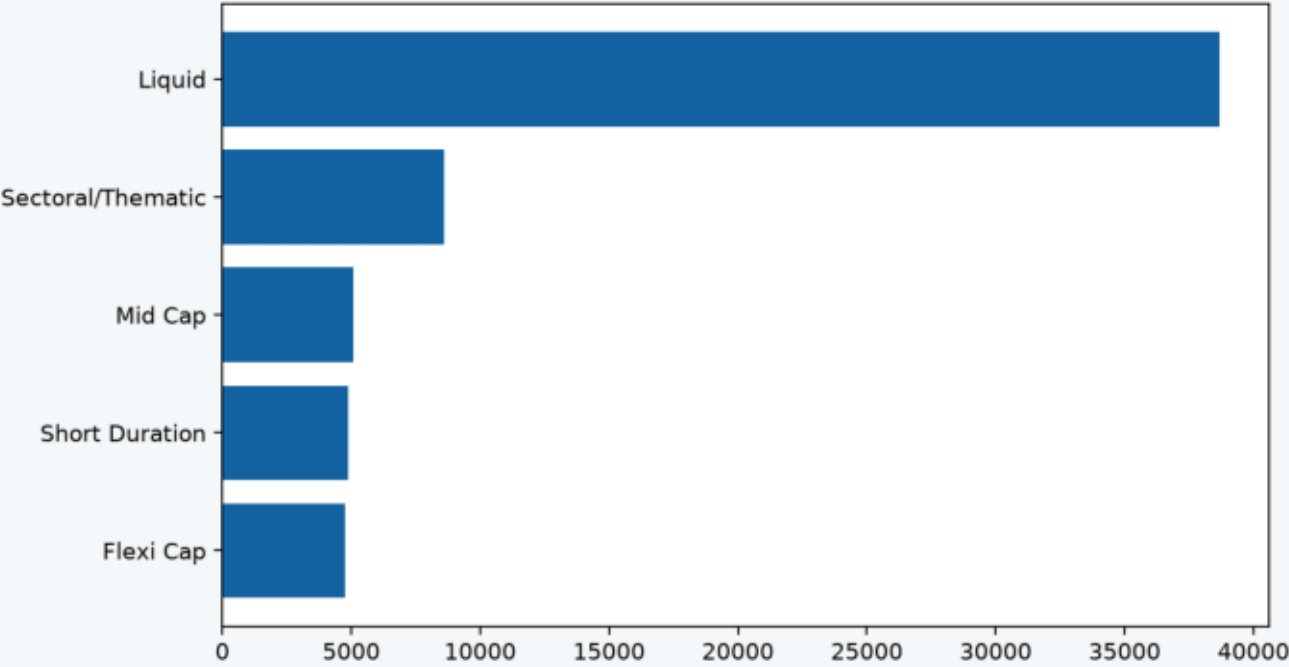
SIP inflows and NIFTY 50



Category inflow heatmap



Top 5 FY25 categories



Limitations

The supplied data is historical and may be synthetic or incomplete. Historical returns, VaR, and rankings do not guarantee future performance.

A native Power BI binary is not generated by Python. The Streamlit app is the interactive alternative; Power BI Desktop can import the cleaned CSV model for a native PBIX implementation.

Recommendations

Use the dashboard filters to compare fund houses, categories, plans, investor cohorts, and flow periods.

Add a scheduled production data refresh, explicit holiday calendars for live NAV ingestion, and governance review before using recommendations in an investor-facing workflow.

Reproducibility

Run: `python scripts/run_pipeline.py`

The runner rebuilds processed CSVs, metrics, SQLite, dashboard exports, and this report/deck. SQLite files are ignored by Git; `sql/schema.sql` is the shareable database definition.

Appendix: Deliverables

ETL and analytics scripts are in scripts/. Notebooks are in notebooks/. SQL definitions are in sql/. Dashboard exports are in bluestock_mf_dashboard_project/. Final report and presentation are in reports/.